

ANTON BREDENBECK

Ph.D. Researcher & Full-Stack Robotician

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EXPERIENCE

Ph.D. Researcher [TU Delft](#)

Supervisors [Salua Hamaza](#) & [Cosimo Della Santina](#)

📅 June 2022 – now 📍 Delft, Netherlands

Projects:

- Developed, implemented, and validated a novel tactile perching approach for aerial robots [1].
- Co-devised a drift-free localization method for tethered drones using only tether-state measurements, with an M.Sc. student [3].
- Visiting researcher @UC Berkeley with [Mark W. Mueller](#): developed tactile high-speed collision recovery and contact-based navigation for UAVs [5].
- Developed compliant-touch contact-force estimation enabling drones to navigate along obstacles [4].
- Devised rolling-contact odometry estimation for drones in contact with surfaces [8].

Academic Roles:

- TA for “Physical Interaction for Aerial and Space Robots”. Designed and conducted exercises, tutorials, exams, and guest lectures.
- Supervisor of B.Sc. (2) and M.Sc. (4) thesis students.
- Member of the [Ph.D. council](#) at the AE Faculty (2024–2026).
- Reviewer: T-RO, T-FR, T-MECH, RA-L, ICRA/IROS, iSpaRo, ICUAS

Research & Teaching Assistant [Julius-Maximilians University](#)

📅 June 2020 – August 2021 📍 Würzburg, Germany

- Designed and implemented the hardware and software of the robotic prototype for 3D point-cloud mapping of lunar caves in the project [DAEDALUS](#).
- TA for “Advanced Sensory Systems and Sensor Data Processing”.

STRENGTHS

[Personal](#) [Hard-working](#) [Resilient](#) [Thorough](#) [Structured](#)

[Robotics](#) [Estimation & Control](#) [Planning](#) [Tactile Sensing](#)

[Mechanical Design](#) [Prototyping](#) [Trajectory Optimization](#)

[Coding](#) [C/C++](#) [Python](#) [ROS 2](#) [ROS](#) [JAX](#) [Julia](#)

REFEREES

Prof. Dr. Salua Hamaza

[@](#) Biomorphic Intelligence Lab,
Associate Professor, TU Delft
✉ s.hamaza@tudelft.nl

Prof. Dr. Mark W. Mueller

[@](#) HiPeR Lab,
Associate Professor, UC Berkeley
✉ mwm@berkeley.edu

Prof. Dr. Cosimo Della Santina

[@](#) Cognitive Robotics,
Associate Professor, TU Delft
✉ c.dellasantina@tudelft.nl

EDUCATION

Ph.D. in Robotics

[Biomorphic Intelligence Lab, TU Delft](#)

📅 June 2022 – now 📍 Delft, Netherlands

I research “Tactile-driven Control for Aerial Robots”. The aim is to enable **aerial robots** to robustly and safely **interact** with the world around them, while exploiting contact as a source of information.

M.Sc. in Satellite Technology

[Julius-Maximilians University](#)

📅 2019 – 2022 📍 Würzburg, Germany

- Graduated with honors (grade 1.0).
- Master’s thesis @ESA/ESTEC working on trajectory optimization and control for a free-floating platform [12].
- Six-month specialization in Robotics & Control @CTU in Prague.

Various Part-Time Jobs:

- Working Student @Zentrum für Telematik e.V. in Würzburg, Germany, designing, manufacturing, and validating electronics for CubeSats.
- Working Student @DFKI in Bremen, Germany, developing a ROS 2 interface to enable reinforcement learning of TurtleBots in the project [Fast & Slow](#).

B.Sc. in Aerospace Computer Science

[Julius-Maximilians University](#)

📅 2016 – 2019 📍 Würzburg, Germany

- Six-month internship @Finnish Geospatial Research Institute in Helsinki, Finland.
- Developed the GNSS interference monitoring service [GNSS-Finland](#) [18] and a dual-frequency measurement correction system for Android devices.

PUBLICATIONS

Journal Articles

- [1] A. Bredenbeck, A. Jadoenathmisier, and S. Hamaza, “Aerial tactile perching via an anthropomorphic hand with embodied soft tactile receptors,” *npj Robotics*, vol. 4, no. 1, p. 35, 2026. DOI: 10.1038/s44182-026-00109-9.
- [2] D. Chaikalis, A. Bredenbeck, M. Brummelhuis, and S. Hamaza, “Sequential multi-tasking aerial manipulation with octopus-inspired tactile tentacles,” *[under review] Nature Communications*, 2026.
- [3] D. van Loon, A. Bredenbeck, L. Puck, M. Azkarate, and S. Hamaza, “Tether-inertial localization for planetary drones,” *IEEE Robotics and Automation Letters*, vol. 11, no. 10, 2026. DOI: 10.1109/LRA.2026.3723306.
- [4] A. Bredenbeck, C. Della Santina, and S. Hamaza, “Embodying compliant touch on drones for aerial tactile navigation,” *IEEE Robotics and Automation Letters*, vol. 10, no. 2, pp. 1209–1216, 2025. DOI: 10.1109/LRA.2024.3519888.
- [5] A. Bredenbeck, T. Yang, S. Hamaza, and M. W. Mueller, “A tactile feedback approach to path recovery after high-speed impacts for collision-resilient drones,” *Drones*, vol. 9, no. 11, 2025. DOI: 10.3390/drones9110758.
- [6] F. Arzberger, J. Zevering, A. Bredenbeck, D. Borrmann, and A. Nüchter, “Unconventional trajectories for mobile 3D scanning and mapping,” *Autonomous Mobile Mapping Robots*, 2022.
- [7] F. Arzberger, A. Bredenbeck, J. Zevering, D. Borrmann, and A. Nüchter, “Towards spherical robots for mobile mapping in human made environments,” *ISPRS Open Journal of Photogrammetry and Remote Sensing*, p. 100 004, 2021.

Conference Proceedings

- [8] M. Schuster*, A. Bredenbeck*, M. Beitelschmidt, and S. Hamaza, “Tactile odometry in aerial physical interaction,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024)*, 2024.
- [9] F. Arzberger, J. Zevering, A. Bredenbeck, D. Borrmann, and A. Nüchter, “Mobile 3D scanning and mapping for freely rotating and vertically descended LiDAR,” in *2022 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, IEEE, 2022, pp. 122–129.
- [10] A. Bredenbeck, C. Della Santina, and S. Hamaza, “End-effector contact force estimation for aerial manipulators,” in *IROS 2022 Workshop on Mobile Manipulation and Embodied Intelligence (MOMA): Challenges and Opportunities*, 2022.
- [11] A. Bredenbeck, S. Vyas, W. Suter, et al., “Finding and following optimal trajectories for an overactuated floating robotic platform,” in *16th Symposium on Advanced Space Technologies in Robotics and Automation (ASTRA)*, 2022.
- [12] A. Bredenbeck, S. Vyas, M. Zwick, D. Borrmann, M. Olivares-Mendez, and A. Nüchter, “Trajectory optimization and following for a three degrees of freedom overactuated floating platform,” in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2022, pp. 4084–4091.
- [13] J. Zevering, D. Borrmann, A. Bredenbeck, and A. Nüchter, “The concept of rod-driven locomotion for spherical lunar exploration robots,” in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2022, pp. 5656–5663.
- [14] D. Borrmann, A. Nüchter, A. Bredenbeck, et al., “Lunar caves exploration with the DAEDALUS spherical robot,” in *52nd Lunar and Planetary Science Conference*, 2021, p. 2073.
- [15] R. Pozzobon, A. Rossi, S. Ferrari, et al., “Marius Hills Skylight hazard characterization as a possible landing site for lunar subsurface exploration,” in *52nd Lunar and Planetary Science Conference*, 2021, p. 1886.
- [16] A. P. Rossi, F. Maurelli, V. Unnithan, et al., “DAEDALUS—descent and exploration in deep autonomy of lava underground structures: Open Space Innovation Platform (OSIP) lunar caves-system study,” 2021.
- [17] J. Zevering, A. Bredenbeck, F. Arzberger, D. Borrmann, and A. Nüchter, “IMU-based pose-estimation for spherical robots with limited resources,” in *2021 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, IEEE, 2021, pp. 1–8.
- [18] S. Nikolskiy, A. Bredenbeck, T. Rikkinen, et al., “GNSS signal quality monitoring based on a reference station network,” in *2020 European Navigation Conference (ENC)*, IEEE, 2020, pp. 1–10.
- [19] J. Zevering, A. Bredenbeck, F. Arzberger, D. Borrmann, and A. Nüchter, “LUNA—a laser-mapping unidirectional navigation actuator,” in *International Symposium on Experimental Robotics*, Springer, Cham, 2020, pp. 85–94.

GRANTS

-  *DIGIT CFP*, approximately USD 1,500, awarded in the form of four GelSight DIGIT sensors for research on tactile control of aerial robots, Meta AI, 2024.
-  *North America Travel Grant*, EUR 4,500 awarded to selected TU Delft AI Labs PhD candidates to support research visits in North America. Used for a research visit to UC Berkeley to work with Mark W. Mueller, AI Lab Initiative, TU Delft, 2024.